

TIME SERIES ANALYSIS & CLASSIFICATION

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Chapter IV: Models for Nonstationary Time Series

1. Introduction
2. Differencing Trend Models
3. Autoregressive Integrated Moving Average (*ARIMA*) Models
4. Variance Stabilizing Transformation

INTRODUCTION

Recall

Suppose $\{e_t\}$ is a zero mean white noise process with variance $\text{var}(e_t) = \sigma_e^2$. In the last chapter, we considered the class of *ARMA* models

$$Y_t = \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \dots + \phi_p Y_{t-p} + e_t - \theta_1 e_{t-1} - \theta_2 e_{t-2} - \dots - \theta_q e_{t-q},$$

or, expressed more succinctly,

$$\Phi(B) Y_t = \Theta(B) e_t,$$

where the *AR* and *MA* characteristic polynomials are

$$\Phi(B) = (1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p),$$

$$\Theta(B) = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q).$$

Recall

- ▶ A process $\{Y_t\}$ in this class is stationary if and only if the roots of the AR characteristic polynomial $\Phi(x)$ all exceed 1 in absolute value (or modulus).
- ▶ A process $\{Y_t\}$ in this class is invertible if and only if the roots of the MA characteristic polynomial $\Theta(x)$ all exceed 1 in absolute value (or modulus).

Introduction

- ▶ In this chapter, we extend this class of models to handle processes which are non-stationary.
- ▶ We accomplish this by generalizing the class of *ARMA* models to include differencing. Doing so gives rise to a much larger class of models, the autoregressive integrated moving average (*ARIMA*) class.
- ▶ This class incorporates a wide range of nonstationary time series processes.

Differencing

- ▶ Suppose that $\{Y_t\}$ is a stochastic process. The first difference process $\{\nabla Y_t\}$ consists of

$$\nabla Y_t = Y_t - Y_{t-1}.$$

- ▶ The second difference process $\{\nabla^2 Y_t\}$ consists of

$$\begin{aligned}\nabla^2 Y_t &= \nabla(\nabla Y_t) = \nabla Y_t - \nabla Y_{t-1} = (Y_t - Y_{t-1}) - (Y_{t-1} - Y_{t-2}) \\ &= Y_t - 2Y_{t-1} + Y_{t-2},\end{aligned}$$

- ▶ In general, the d th difference process $\{\nabla^d Y_t\}$ consists of

$$\nabla^d Y_t = \nabla(\nabla^{d-1} Y_t) = \nabla^{d-1} Y_t - \nabla^{d-1} Y_{t-1},$$

for $d = 1, 2, \dots$ etc. We take $\nabla^0 Y_t = Y_t$ by convention.

Differencing: Example

- ▶ Suppose that $\{Y_t\}$ is a random walk process

$$Y_t = Y_{t-1} + e_t,$$

where $\{e_t\}$ is zero mean white noise with variance $\text{var}(e_t) = \sigma_e^2$. We know that $\{Y_t\}$ is not stationary because its autocovariance function depends on t (see Chapter 2).

- ▶ However, the first difference process

$$\nabla Y_t = Y_t - Y_{t-1} = e_t$$

is white noise, which is stationary.

DIFFERENCING TREND MODELS

Differencing Trend Models

- We talked about how to use regression methods to fit models of the form

$$Y_t = \mu_t + X_t,$$

where μ_t is a deterministic trend function and where $\{X_t\}$ is a stochastic process with $E(X_t) = 0$.

Differencing Trend Models

- Suppose that $\{X_t\}$ is stationary and that the true trend function is

$$\mu_t = \beta_0 + \beta_1 t,$$

a linear function of time. Clearly, $\{Y_t\}$ is not a stationary process because

$$E(Y_t) = E(\beta_0 + \beta_1 t + X_t) = \beta_0 + \beta_1 t + E(X_t) = \beta_0 + \beta_1 t,$$

which depends on t .

Differencing Trend Models

- ▶ The first differences are given by

$$\nabla Y_t = Y_t - Y_{t-1} = (\beta_1 + X_t - X_{t-1}).$$

- ▶ Note that

$$E(\nabla Y_t) = E(\beta_1 + X_t - X_{t-1}) = \beta_1 + E(X_t) - E(X_{t-1}) = \beta_1.$$

Also,

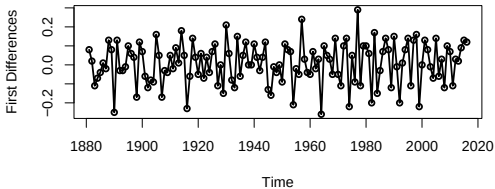
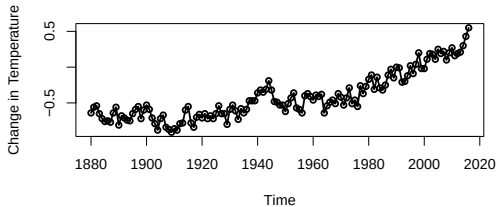
$$\text{Cov}(\nabla Y_t, \nabla Y_{t-k}) = \text{Cov}(\beta_1 + X_t - X_{t-1}, \beta_1 + X_{t-k} - X_{t-k-1})$$

Because $\{X_t\}$ is stationary, each of the cross covariance terms does not depend on t .

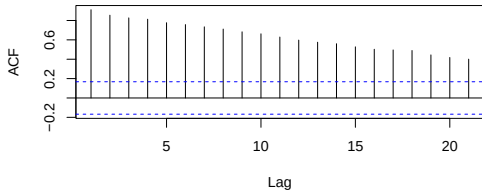
Differencing Trend Models

- ▶ Therefore, both $E(\nabla Y_t)$ and $\text{Cov}(\nabla Y_t, \nabla Y_{t-k})$ are free of t ; i.e., $\{\nabla Y_t\}$ is a stationary process.
- ▶ Taking first differences removes a linear deterministic trend.

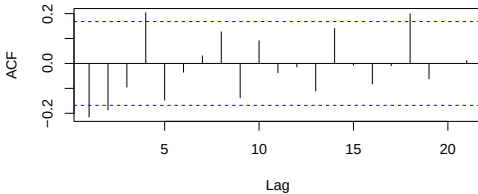
Differencing Trend Models



ACF of the process



ACF of the differenced process



Differencing Trend Models: Generalisation

- Suppose that $Y_t = \mu_t + X_t$, where μ_t is a deterministic trend function and $\{X_t\}$ is a stationary process with $E(X_t) = 0$. In general, if

$$\mu_t = \beta_0 + \beta_1 t + \beta_2 t^2 + \dots + \beta_d t^d$$

is a polynomial in t of degree d , then the d^{th} difference process $\{\nabla^d Y_t\}$ is stationary.

AUTOREGRESSIVE INTEGRATED MOVING AVERAGE (*ARIMA*) MODELS

Terminology

- ▶ A stochastic process $\{Y_t\}$ is said to follow an *ARIMA* model if the d th differences $W_t = \nabla^d Y_t$ follow a stationary *ARMA* model.
 - p : Order of the autoregressive component
 - d : Number of differences needed for a stationary $ARMA(p, q)$ process
 - q : Order of the moving average component
- ▶ **Relationship:** Y_t is $ARIMA(p, d, q) \Leftrightarrow W_t = \nabla^d Y_t$ is $ARMA(p, q)$.

Recall

- ▶ A stationary $ARMA(p, q)$ process:

$$(1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p) Y_t = (1 - \theta_1 B - \theta_2 B^2 - \dots - \theta_q B^q) e_t$$
$$\Phi(B) Y_t = \Theta(B) e_t,$$

where $\{e_t\}$ is zero mean white noise, $\text{Var}(e_t) = \sigma_e^2$.

- ▶ In general, $ARIMA(p, d, q)$: $\Phi(B)(1 - B)^d Y_t = \Theta(B) e_t$.

Important

- ▶ In practice, $d > 2$ is rarely needed. Most real-time series data can be coerced into a stationary *ARMA* process by taking one or two differences.

Remark

AR, *MA*, and *ARMA* models are all members of the *ARIMA* family:

- ▶ $AR(p) \longleftrightarrow ARIMA(p, 0, 0)$
- ▶ $MA(q) \longleftrightarrow ARIMA(0, 0, q)$
- ▶ $ARMA(p, q) \longleftrightarrow ARIMA(p, 0, q)$
- ▶ $ARI(p, d) \longleftrightarrow ARIMA(p, d, 0)$
- ▶ $IMA(d, q) \longleftrightarrow ARIMA(0, d, q)$

Example

- Suppose $\{e_t\}$ is a zero mean white noise process. Identify the model:

$$Y_t = 1.5 Y_{t-1} - 0.5 Y_{t-2} + e_t - e_{t-1} + 0.25 e_{t-2}$$

as an $ARIMA(p, d, q)$ process. That is, specify the values of p , d , and q .

- Upon first glance, $Y_t = 1.5 Y_{t-1} - 0.5 Y_{t-2} + e_t - e_{t-1} + 0.25 e_{t-2}$ looks like an $ARMA(2, 2)$ process, but this process is not stationary.

Example

- To see why, note that we can write this process as

$$\begin{aligned} Y_t - 1.5Y_{t-1} + 0.5Y_{t-2} &= e_t - e_{t-1} + 0.25e_{t-2} \\ \Rightarrow (1 - 1.5B + 0.5B^2)Y_t &= (1 - B + 0.25B^2)e_t \\ \Rightarrow (1 - 0.5B)(1 - B)Y_t &= (1 - 0.5B)^2e_t \\ \Rightarrow (1 - B)Y_t &= (1 - 0.5B)e_t \\ \Rightarrow W_t &= (1 - 0.5B)e_t, \end{aligned}$$

where $W_t = (1 - B)Y_t = Y_t - Y_{t-1}$.

- Here, the first differences $\{W_t\}$ follow an $MA(1)$ model with $\theta = 0.5$.
Therefore, $\{Y_t\}$ is an $ARIMA(0, 1, 1) \Leftrightarrow IMA(1, 1)$ process with $\theta = 0.5$.

VARIANCE STABILIZING TRANSFORMATION

Transformations: Overview

- ▶ If there is clear evidence of non constant variance over time (e.g., the variance increases over time, etc.), then a suitable transformation to the data might remove (or lessen the impact of) the non constant variance pattern
- ▶ Applying a transformation to address non constant variance is regarded as a **first step**. This is done before using differencing as a means to achieve stationarity.

Variance Stabilizing Transformation

- ▶ Suppose that the variance of non stationary process $\{Y_t\}$ can be written as $\text{Var}(Y_t) = c_0 f(\mu_t)$, where $\mu_t = E(Y_t)$ and c_0 is a positive constant.
- ▶ The variance is not constant because it is a function of μ_t , which is changing over time.
- ▶ Our goal is to find a function T so that the transformed series $T(Y_t)$ has constant variance:

$$\text{Var}[T(Y_t)] \approx c_1,$$

where c_1 is a constant.

Variance Stabilizing Transformation

- Consider approximating the function T by a first-order Taylor-series expansion about the point μ_t , that is,

$$T(Y_t) \approx T(\mu_t) + T'(\mu_t)(Y_t - \mu_t),$$

where $T'(\mu_t)$ is the first derivative of $T(Y_t)$, evaluated at μ_t .

- Note that

$$\text{Var}[T(Y_t)] \approx \text{Var}[T(\mu_t) + T'(\mu_t)(Y_t - \mu_t)] = c_0 [T'(\mu_t)]^2 f(\mu_t).$$

- Therefore, we want to find the function T which satisfies

$$\text{Var}[T(Y_t)] \approx c_0 [T'(\mu_t)]^2 f(\mu_t) = c_1,$$

where c_1 is a constant free of μ_t .

Variance Stabilizing Transformation

- Solving this expression for $T'(\mu_t)$, we get the differential equation

$$T'(\mu_t) = \sqrt{\frac{c_1}{c_0 f(\mu_t)}} = \frac{c_2}{\sqrt{f(\mu_t)}},$$

where $c_2 = \sqrt{\frac{c_1}{c_0}}$ is free of μ_t . Integrating both sides, we get

$$T(\mu_t) = \int \frac{c_2}{\sqrt{f(\mu_t)}} d\mu_t + c_3,$$

where c_3 is a constant free of μ_t .

Variance Stabilizing Transformation

Examples:

- If $\text{var}(Y_t) = c_0\mu_t$, then

$$T(\mu_t) = \int \frac{c_2}{\sqrt{\mu_t} d\mu_t} = 2c_2\sqrt{\mu_t} + c_3$$

Taking $c_2 = 1/2$ and $c_3 = 0$ we see that the square root of the series, $T(Y_t) = \sqrt{Y_t}$, will provide a constant variance.

Variance Stabilizing Transformation

Examples:

- If $\text{var}(Y_t) = c_0\mu_t^2$, then $T(Y_t) = \ln(Y_t)$.
- If $\text{var}(Y_t) = c_0\mu_t^4$, then $T(Y_t) = \frac{1}{Y_t}$.

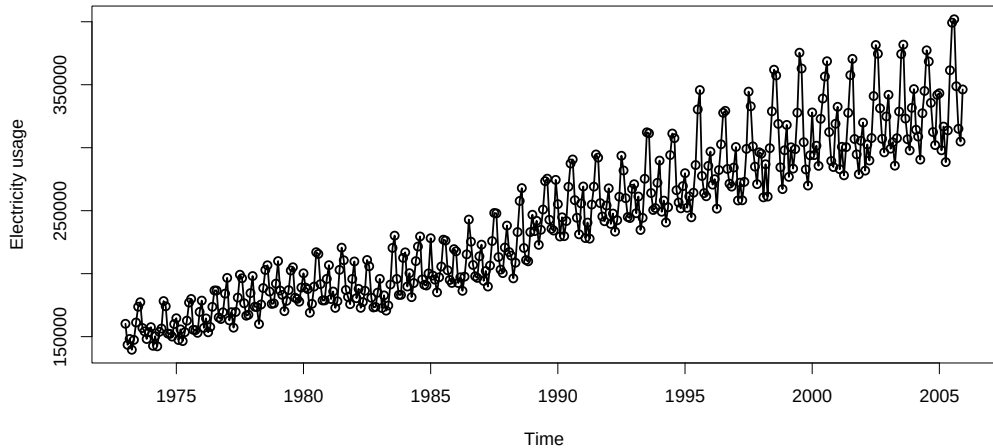
► **Box-Cox Transformations:** More generally, a flexible family of transformations is given by:

$$T(Y_t) = \begin{cases} \frac{Y_t^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln(Y_t), & \lambda = 0 \end{cases}$$

Box-Cox Transformations

λ	$T(Y_t)$	Description
-2.0	$-\frac{1}{Y_t^2}$	Inverse square
-1.0	$-\frac{1}{Y_t}$	Reciprocal
-0.5	$\frac{1}{\sqrt{Y_t}}$	Inverse square root
0.0	$\ln(Y_t)$	Logarithm
0.5	$\sqrt{Y_t}$	Square root
1.0	Y_t	Identity (no transformation)
2.0	Y_t^2	Square

Example



Example

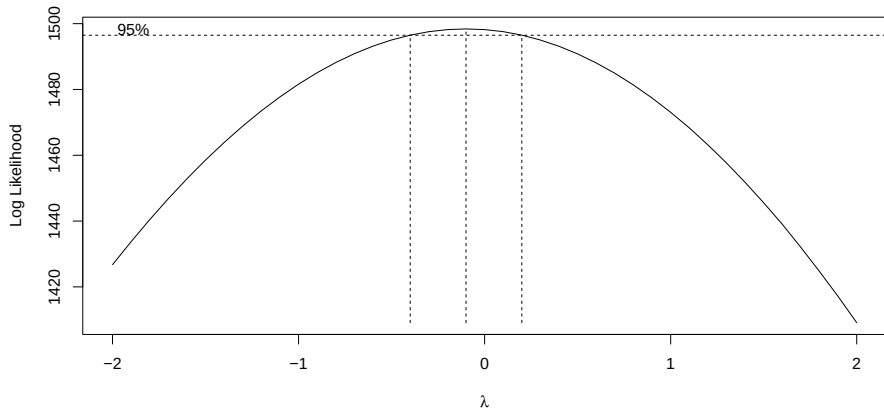


Fig. Use the function `BoxCox.ar(electricity)`!, $\hat{\lambda} = -0.1$, $CI = [-0.4 \ 0.2]$

Example

